

Finite-Size Majorana Splitting in the Kitaev Chain: Why do the “spikes” in the finite-size spectrum worsen with larger L ?

Seminar notes

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1 Setup

We study the Kitaev chain at $t = \Delta = 1$ with open boundary conditions (OBC). The BdG Hamiltonian is the real-symmetric matrix

$$M = \begin{pmatrix} h & d \\ -d & -h \end{pmatrix}, \quad h_{ij} = -\mu\delta_{ij} - t(\delta_{i,j+1} + \delta_{i+1,j}), \quad d_{ij} = \Delta(\delta_{i,j+1} - \delta_{i+1,j}). \quad (1)$$

Particle-hole symmetry (PHS) guarantees eigenvalues come in exact $\pm E$ pairs. Plot 4 tracks the *six smallest positive eigenvalues* as μ varies.

2 The ideal edge-mode energy

2.1 Zero-energy condition in the complex BZ

To find the edge-mode energy, we look for a zero-energy solution that decays into the bulk. Set $E = 0$ and try an Ansatz $\psi_j \sim \lambda^j$. Substituting into the bulk BdG equations gives the characteristic equation

$$(-\mu - t(\lambda + \lambda^{-1}))^2 + \Delta^2(\lambda - \lambda^{-1})^2 = 0. \quad (2)$$

For $t = \Delta$ this factors to

$$(-\mu - t(\lambda + \lambda^{-1}) + \Delta(\lambda - \lambda^{-1}))(-\mu - t(\lambda + \lambda^{-1}) - \Delta(\lambda - \lambda^{-1})) = 0. \quad (3)$$

The two branches give the solutions

$$\lambda_A = \frac{\mu}{2t}, \quad \lambda_B = \frac{2t}{\mu}. \quad (4)$$

For the left edge mode we need $|\lambda| < 1$, so we take $\lambda = \mu/(2t)$ (valid for $|\mu| < 2t$, i.e. inside the topological phase). The wavefunction decays *purely exponentially*:

$$\psi_j^L \sim \left(\frac{|\mu|}{2t}\right)^{j-1}, \quad \xi = \frac{1}{|\ln |\mu/2t||} \quad (\text{coherence length}). \quad (5)$$

Crucially, for $t = \Delta$ the solution has **no oscillatory factor** e^{ik_0j} . This is special to the symmetric point; for $t \neq \Delta$ the roots can be complex, introducing oscillations.

2.2 Finite-size hybridization splitting

A chain of length L has both a left edge mode $\psi_j^L \sim \lambda^{j-1}$ and a right edge mode $\psi_j^R \sim \lambda^{L-j}$. Their overlap determines the splitting of the zero-energy pair:

$$\delta E \approx 2t \frac{\langle \psi^L | V | \psi^R \rangle}{\|\psi^L\| \|\psi^R\|} \sim 2t \lambda^L = 2t \left(\frac{|\mu|}{2t} \right)^L. \quad (6)$$

This is the energy of the near-zero mode. For $t = \Delta = 1$:

$$E_0(\mu, L) \approx 2 \left(\frac{|\mu|}{2} \right)^L. \quad (7)$$

Predictions from (7):

- At the sweet spot $\mu = 0$: $E_0 = 0$ *exactly*, for any L .
- At the phase boundary $|\mu| = 2t$: $E_0 \rightarrow 2$ (the bulk gap), for any L .
- In the interior: E_0 decreases *exponentially* with L . A larger chain has a better-protected zero mode.

So (7) predicts that larger L should give *smaller* spikes, not larger ones.

3 Why the spikes increase and widen with L : a numerical artifact

3.1 The floating-point threshold problem

The observed behaviour (spikes worsen as L increases) is *not* a physical effect — it is a numerical artefact arising from how we define “positive spectrum”.

In the simulation, positive eigenvalues are selected with the filter `evals[evals >= 0]`. For large L , the ideal splitting $E_0 = 2(|\mu|/2)^L$ falls below machine precision ($\approx 10^{-15}$) over a wide range of μ :

$$E_0 < \varepsilon_{\text{mach}} \iff \left(\frac{|\mu|}{2} \right)^L < \frac{\varepsilon_{\text{mach}}}{2} \iff |\mu| < 2\varepsilon_{\text{mach}}^{1/L}. \quad (8)$$

For $\varepsilon_{\text{mach}} = 5 \times 10^{-16}$:

$$L = 30 : \quad |\mu|_{\text{thresh}} \approx 0.62, \quad (9)$$

$$L = 100 : \quad |\mu|_{\text{thresh}} \approx 1.41. \quad (10)$$

For $|\mu| < |\mu|_{\text{thresh}}$, the near-zero eigenvalue is indistinguishable from zero at double precision. Due to rounding, it can be stored as a tiny *negative* number ($\sim -10^{-16}$). The filter `evals >= 0` then *excludes* this eigenvalue, and the *lowest bulk mode* (at finite energy $\sim E_{\text{gap}}$) takes its place as the “lowest positive eigenvalue”. This appears as a spike.

3.2 Width and height of the artefact

- **Width:** the spike region is $|\mu| < 2\varepsilon_{\text{mach}}^{1/L}$, which *grows* with L (for $L = 100$ it covers most of the topological phase, explaining the “widened” spikes).
- **Height:** the spike jumps to the lowest bulk energy, which is $\sim E_{\text{gap}}(\mu)/(L\text{-dependent level spacing})$. With larger L the bulk level spacing is smaller, so more bulk levels fall inside the window and the spikes look denser.

4 The fix: use quasiparticle energies $|E|$

The physical quasiparticle spectrum consists of the *absolute values* of the BdG eigenvalues. The near-zero mode has $|E| \approx 0$ regardless of the sign of the floating-point result:

$$\text{correct : } \{E_n\}_{n=1}^L = \text{one copy from each adjacent pair in sort}(|e_i|). \quad (11)$$

In code, replace

```
return evals[evals >= 0]          # WRONG: sign-sensitive
```

with

```
return np.sort(np.abs(evals))[:,2] # correct: |E| plus pair deduplication
```

The absolute value is what protects the near-zero Majorana mode from a floating-point sign flip. The stride is what removes the particle-hole double counting: each physical quasiparticle energy appears twice as $\pm E$ before taking the absolute value.

5 Summary

Observation	Physical or artefact?	Cause
Spikes near $ \mu \rightarrow 2t$	Physical	$\xi \rightarrow \infty$, hybridisation is strong
Spikes in interior for small L	Mostly artefact	Float precision, level crossings
Spikes widen/worsen for large L	Artefact	Threshold $ \mu _{\text{thresh}} \propto \varepsilon^{1/L}$ grows